

Embedded Systems & Cortex-M4 Architecture

Microcontrollers vs. Microprocessors

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Agenda

What is an Embedded System?

Our Board: Nucleo-64

MCU Cortex-M4 Architecture

Closing

What is an Embedded System?

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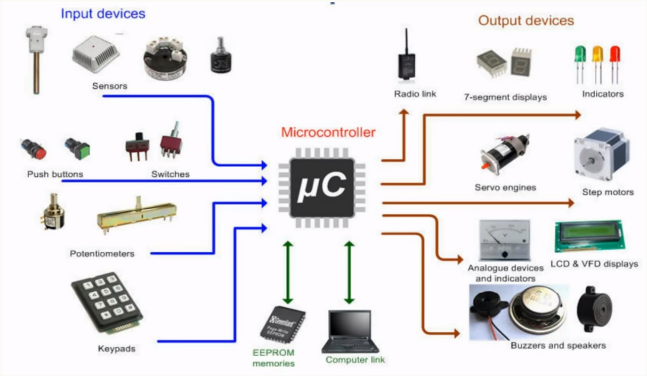


Figure 1: Embedded system.[1]

[1] Original image from *www.electroniclinic.com*

What is an Embedded System?

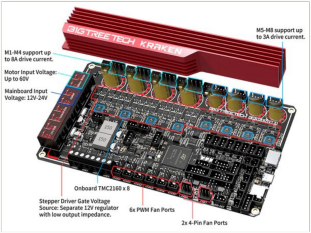
- An embedded system is a combination of computer hardware and software designed for a specific function.
- It is “embedded” as part of a complete device, often including mechanical parts.
- It is *not* a general-purpose computer like a laptop — you can’t install new apps on your microwave!
- Key characteristic: it performs a dedicated, pre-defined task.

Applications for Embedded Systems









The Brain of the System: MPU vs. MCU





Microprocessor (MPU) — A Powerful Brain

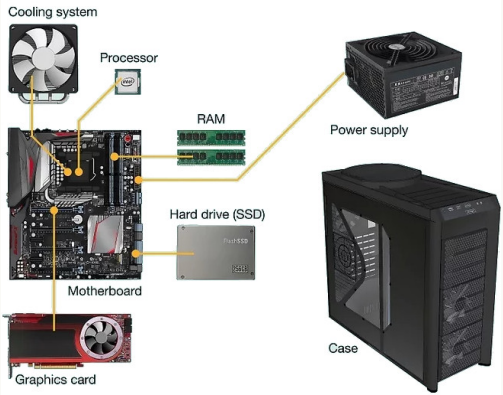


Figure 2: Desktop PC.[2]

[2] Original image from *www.wikihow.com*

Microcontroller (MCU) — A Computer on a Chip

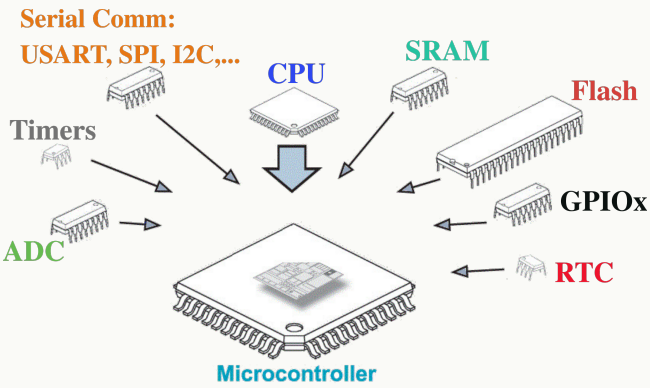


Figure 3: MCU general idea.[3]

[3] Original image from *www.electronicshub.org*

What is a Microcontroller (MCU)?

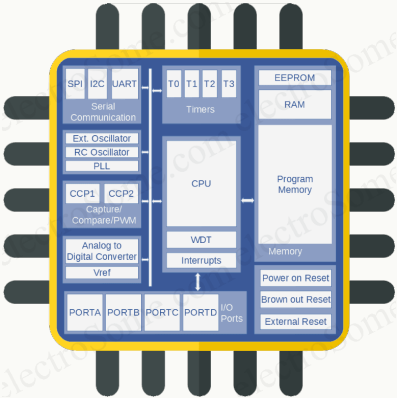
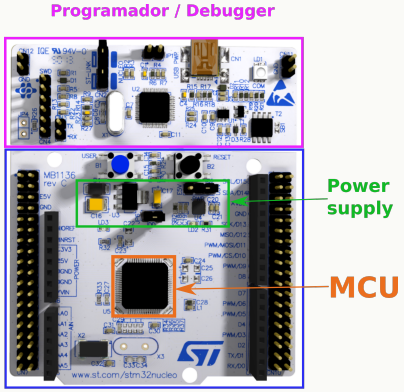


Figure 4: MCU general idea.[4]

[4] Original image from *electrosome.com*

Our Board: Nucleo-64

Why We Use an MCU for This Course



MCUs are the heart of the vast majority of embedded products.

- They are self-contained: everything we need to control is on one chip.
- They allow direct hardware control: code to directly manipulate physical pins, timers, and sensors.
- They provide real-time feedback: you see the immediate result of your code (an LED blinks, a motor spins).

MCU Cortex-M4 Architecture

How is a Cortex-M4 MCU?

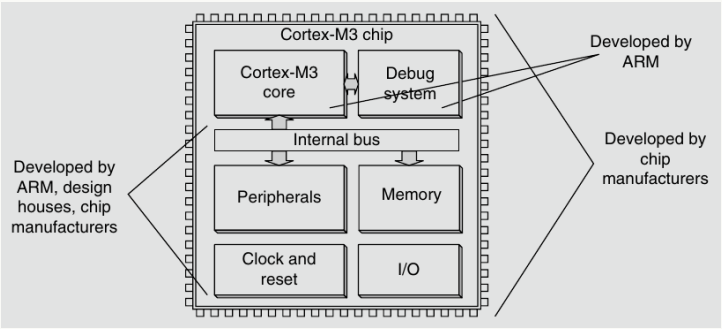


Figure 5: Generic Cortex-M3 MCU.[5]

[5] Original image from book *"The Definitive Guide to ARM Cortex-M3"*

How is a Cortex-M4 MCU?

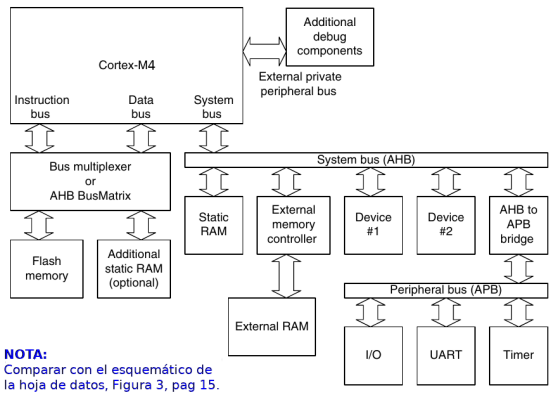


Figure 6: Cortex-M4 detail: buses and peripherals.[6]

[6] Original image from book *"The Definitive Guide to ARM Cortex-M3"*

SFR — Cortex-M4 Memory

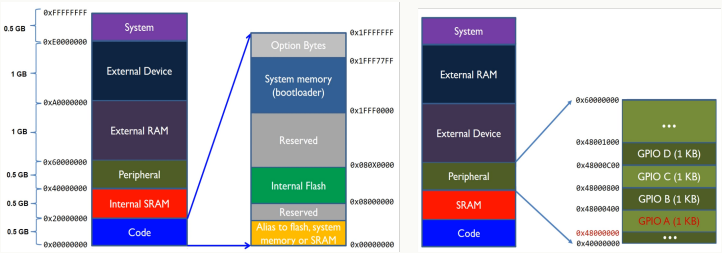


Figure 7: Memory distribution.[7]

[7] Original image from YouTube, *"Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C"*

SFR — Cortex-M4 Memory

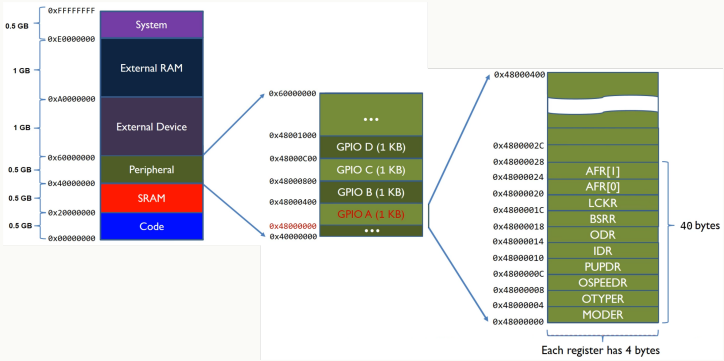


Figure 8: SFR detail of the GPIOx peripheral.[8]

[8] Original image from YouTube, *"Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C"*

Closing

To Think a While...

Applications of an MCU inside a car...

MCUs Inside a Car...

